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Shaping the future of survey statistics in the data-driven era
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Beyond String Matching: Embeddings and Artificial Intelligence for Statistical Integration

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- Artificial Intelligence in Official Statistics
- ATECO Classification
- Automated Coding Based on Deterministic Methods
- Automated Coding Based on Artificial Intelligence
- Conclusion

Non-Traditional Data Sources

Modern AI methods allow for harnessing non-traditional data sources

- **Administrative data:** administrative data sources containing complex texts (e.g., company balance sheets, patents). **Semantic search techniques** allow to **assign classification codes** (NACE, ISCO) to each data source.
- **Company websites:** web scraping combined with **Retrieval Augmented Generation (RAG)** workflows allow the estimate of enterprises' products.
- **Social media data:** tweets are filtered, classified using **sentiment analysis**, and validated through **word embeddings** and **topic modeling** techniques to analyze themes like hate speech.

Semantic Search for Retrieving Statistical Data

Traditional search engines require users to know the **exact keywords** used in table *titles* or *metadata*. This limits accessibility, especially for non-experts unfamiliar with statistical jargon.

Semantic search engines

Semantic search allows users to write natural queries like:

- "companies performing e-commerce"
- "drinking habits of italian population"

The engine interprets the **semantic** meaning of users' queries, improving:

- **relevance of results**
- user experience
- accessibility to a wider audience

Semantic Search for Retrieving Statistical Data

1 Artificial Intelligence in Official Statistics

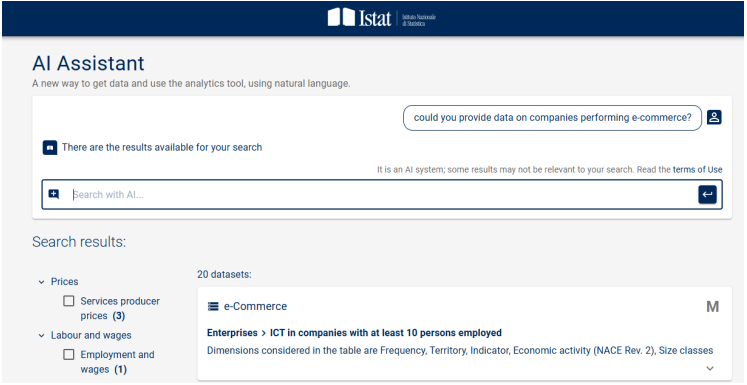

Figure: Screenshot of the AI Assistant integrated in the IstatData portal.

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The Italian version of the NACE classification

2 ATECO Classification

The **ATECO** classification, used in many statistical domains (e.g., business registers, enterprise surveys) has the following characteristics:

Hierarchical, with multiple levels: **sections**, **divisions**, **groups**, **classes**, and **subcategories**.

Semantic-rich, with each code described through:

- **Central notes**: applying to a group and its descendants.
- **Exclusion notes**: clarifying boundaries by specifying what is **not** included.
- **Inclusion notes**: providing examples and clarifications.

This complexity makes automatic classification a challenging task.

Structure and Complexity of ATECO Notes

2 ATECO Classification

01.1 Growing of non-perennial crops

This group includes the growing of non-perennial crops, in other words, plants that do not last for more than two growing seasons. Included is the growing of these plants for the purpose of seed production.

01.11 Growing of cereals, other than rice, leguminous crops and oil seeds

This class includes all forms of growing of cereals, leguminous crops and oil seeds. Further included is the growing of these plants for the purpose of seed production. The growing of these crops is often combined within agricultural units.

The following activities are excluded:

- *growing of rice, see 01.12*
- *growing of vegetables, see 01.13*
- *growing of sweetcorn, see 01.13*
- *growing of fibre crops, see 01.16*
- *growing of maize, lupines or kale for fodder, see 01.19*
- *growing of oleaginous fruits, see 01.26*

01.11.0 Growing of cereals, other than rice, leguminous crops and oil seeds

01.11.00 Growing of cereals, other than rice, leguminous crops and oil seeds

- growing of cereals, e.g.:
  - wheat
  - grain maize
  - sorghum
  - barley

Group

Central Note
located at group 01.1
which applies to code 01.1
and all its children

Class

Exclusion Note
located in class 01.11
which apply to code 01.11
and all its children

Sub-category

Inclusion Note (containing item list)
located in subcategory 01.11.00
which apply to code 01.11.11
and all its parents

▶ Artificial Intelligence in Official Statistics

▶ ATECO Classification

▶ Automated Coding Based on Deterministic Methods

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A Generalized Automated Coding System

3 Automated Coding Based on Deterministic Methods

Wenzowski, a researcher from Statistics Canada, introduced in 1988 a generalized method for automated text classification.

ACTR performs deterministic coding of free-text descriptions by matching them with classification items using:

- **Dictionary**
- **Parsing strategy**
- **Matching strategy**

Survey Methodology, December 1988
Vol. 14, No. 2, pp. 299–307
Statistics Canada

ACTR
A Generalized Automated Coding System

M.J. WENZOWSKI[1]

ABSTRACT

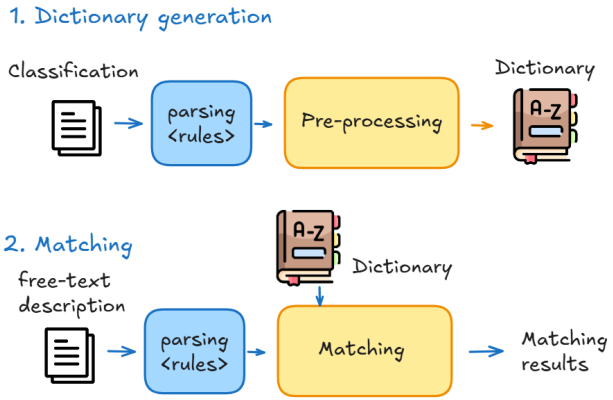
A generalized implementation of a method for performing automated coding is described. Traditionally, coding has been performed manually by specially trained personnel, but recently computerized systems have appeared which either eliminate or substantially reduce the need for manual coding. Typically, such systems are limited in use to those applications for which they were originally designed. The system presented here may be used by any application to perform coding of English or French text using any classification scheme.

KEY WORDS: Automated coding; Classification; Text searching.

Comprehensive Istat R Coding Environment

3 Automated Coding Based on Deterministic Methods

CIRCE, developed by Istat and based on ACTR, is a tool for coding free-text descriptions into official statistical classifications (e.g., NACE, ISCO).



Dictionary Generation

3 Automated Coding Based on Deterministic Methods

1. Remove hierarchical levels

01.11 Growing of cereals, other than rice, leguminous crops and oil seeds

2. Remove non-informative text using regular expressions

01.11 Growing of cereals, other than rice, leguminous crops and oil seeds → 01.11 Growing of cereals, leguminous crops and oil seeds

3. Tokenization and stopwords removal

Growing of cereals, leguminous crops and oil seeds → Growing, cereals, leguminous, crops, oil, seeds

4. Generate bigrams and calculate weights

Growing of cereals, leguminous crops and oil seeds → 1.0 growing&cereals
0.25 growing&crops
...

The pre-processing phase builds an index for each entry in the classification

01.11 Growing of cereals, other than rice, leguminous crops and oil seeds

↓ Index

1.0 growing&cereals
0.25 growing&crops
0.5 growing&seeds
...

Notes on Dictionary Generation

3 Automated Coding Based on Deterministic Methods

- **Expert validation is essential:**
  The dictionary index is generated through preprocessing, but each extracted token (*unigram, bigram, trigram*) must be **reviewed and validated by domain experts to ensure it captures relevant meaning**.
- **Index structure:**
  For each subcategory in the ATECO classification, the dictionary may contain a list of weighted keys representing its semantic signature:

  Example: 01.11.00 → `growing&cereals (1.0), growing&crops (0.25), growing&seeds (0.5)`
- **Maintenance cost:**
  Adding or updating classifications requires full regeneration of the dictionary and a complete review by domain experts.

Matching Phase

The input text is compared against all dictionary entries:

The matching phase calculates a similarity score (S), by comparing the input text with each dictionary entry. Based on the score, *computed as a weighted average on matching terms*, CIRCE determines the most suitable match using the following rules.

Matching outcomes:

$$\begin{aligned}
\text{Unique:} \quad & S_1 > S_{\max} \quad \text{and} \quad (S_1 - S_2) > \Delta S \\
\text{Multiple:} \quad & S_1 > S_{\max} \quad \text{and} \quad (S_1 - S_2) < \Delta S \\
\text{Possible:} \quad & S_{\min} < S_1 \leq S_{\max} \\
\text{Failed:} \quad & S_1 \leq S_{\min} \quad \text{or no } S_1
\end{aligned}$$

The thresholds $S_{min}, S_{max}, \Delta S$ can be tuned by the user.

Limitations of the Classical Approach

- **Parsing strategy requires manual maintenance:**
  Rules for tokenization, normalization, and **replacement words** must be updated to cover spelling variants and synonyms.
- **No semantic understanding:**
  Sentences are reduced to standardized tokens sorted alphabetically.
  E.g., "vendere prodotti agricoli" → vendere, prodotti, agricoli
- **Rigid matching logic:**
  Exact or near-exact word overlap is needed to trigger a match; rephrasings are often missed.
- **Low scalability:**
  Adapting to new domains or tasks requires building new dictionaries and strategies from scratch.

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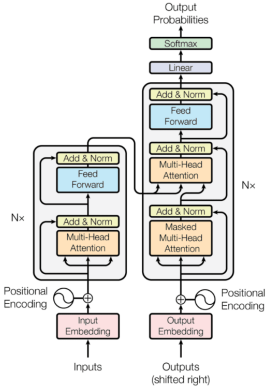
A Breakthrough in AI Text Understanding

Attention is All You Need:

- Introduced the **Transformer architecture**, revolutionizing NLP.
- Based on a new mechanism: **Self-Attention**.
- Enables capturing long-range dependencies and semantic relations.
- Foundation of modern Large Language Models (e.g., BERT, GPT).

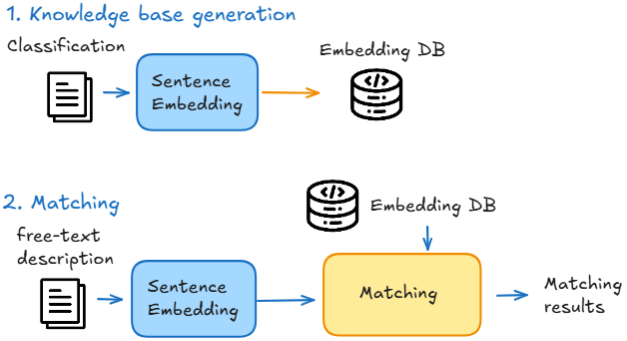


Transformer enables accurate semantic matching of textual descriptions, marking a **shift from rule-based to meaning-based approaches**.

Semantic Search Workflow

4 Automated Coding Based on Artificial Intelligence

To test Artificial Intelligence methods for automated coding, we implemented a semantic search workflow for coding free-text descriptions into official statistical classifications (e.g., NACE, ISCO).



Knowledge Base Generation

4 Automated Coding Based on Artificial Intelligence

We explored several approaches. For our prototype, we settled on the most detailed classification level (5 and 6 digits), where for each code, we create one or more descriptive elements.

01.13.30 Growing of tubers, including potatoes

- growing of tubers, e.g.:
- potatoes
- sweet potatoes
- cassava
- yams
- swedes

→

01.13.30

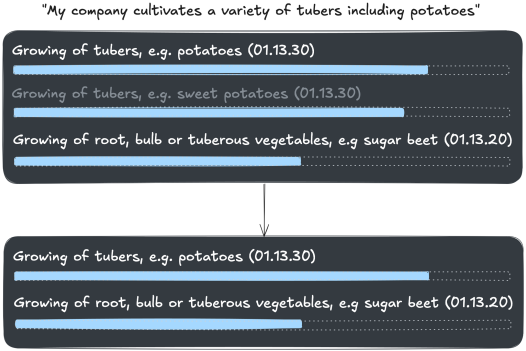
- Growing of tubers, e.g. potatoes.
- Growing of tubers, e.g. sweet potatoes.
- Growing of tubers, e.g. cassava.
- Growing of tubers, e.g. yams.
- Growing of tubers, e.g. swedes.

Each text is embedded using an open sentence embedding model (`BAAI/bge-m3`).

Matching Phase

4 Automated Coding Based on Artificial Intelligence

Once the user inputs a query, the model embeds it and retrieves the top_k matches. If multiple retrieved vectors refer to the same code, only the most similar is retained.



Limitations of the AI Approach

- **High Computational Costs:**
  Embedding-based models can be resource-intensive, especially when applied to large knowledge bases or complex document collections.
- **Representation Biases:**
  Sentence embeddings are influenced by training objectives and may overlook key distinctions. For example, the sentences *"including rice"* and *"excluding rice"* could be mistakenly considered similar, leading to inaccurate retrievals in coding tasks.
- **Domain specificity:**
  Most models are trained on general-purpose text and may underperform on domain-specific content. **Expert-driven fine-tuning and validation are essential to ensure reliable results.**

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Comparison Between Approaches

Classical Approach	Semantic Search
• Static approach with custom hard-coded rules.	• Flexible embedding-based approach that automatically handles semantics.
• Low computational resources.	• High computational resources.
• Scales poorly to new classifications.	• Adapts easily to new classifications.
• Domain expertise is needed to build the vocabulary (heavy workload).	• Domain expertise is needed to validate and improve the model (light workload).

Future Work and Opportunities

- First, we need to **validate** our approach. This will be done in two ways:
  - Comparing our results on the 2022 version of the ATECO classification (the one used by CIRCE) with the baseline.
  - Manually labeling a set of texts produced by enterprises using the 2025 ATECO version and evaluating our system on it.
- We plan to integrate the human-labeled vocabulary that CIRCE uses to fine-tune and specialize our models.
- When our system goes into production, we will carry out **continuous learning** to dynamically improve its performance based on user feedback and activity.
- Domain expertise will be utilized for validating and enhancing the system's performance, rather than building vocabularies.